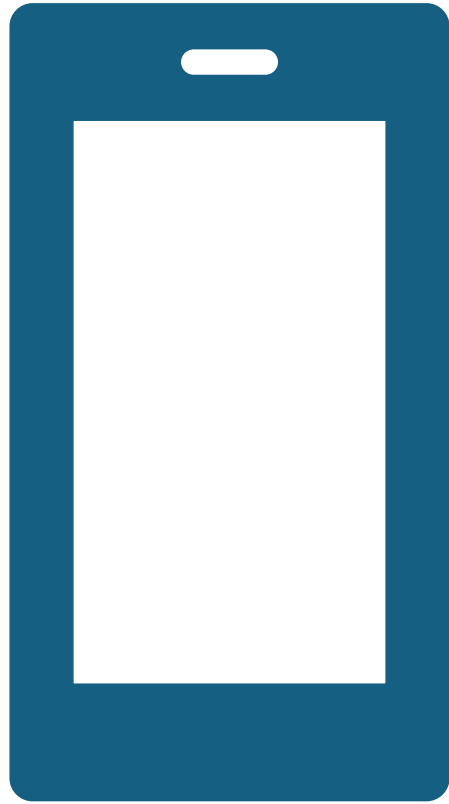




# Smart Device Usage Analysis for Bellabeat

Turning User Behavior Insights into Data-Driven Marketing  
Strategies

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# Agenda



INTRODUCTION &  
PURPOSE



DATA & ANALYSIS



KEY FINDINGS



RECOMMENDATIONS



ACTIONABLE STEPS

# Introduction

- **Problem statement:**

Bellabeat needs to understand when and how users are active during the day, and how activity relates to health metrics (calories, heart rate, sleep) in order to design more effective marketing strategies.

- **Business task:**

The purpose of this analysis is to understand how users interact with smart fitness devices throughout the day and week, and to use these insights to improve Bellabeat's marketing strategy.



# Data Overview

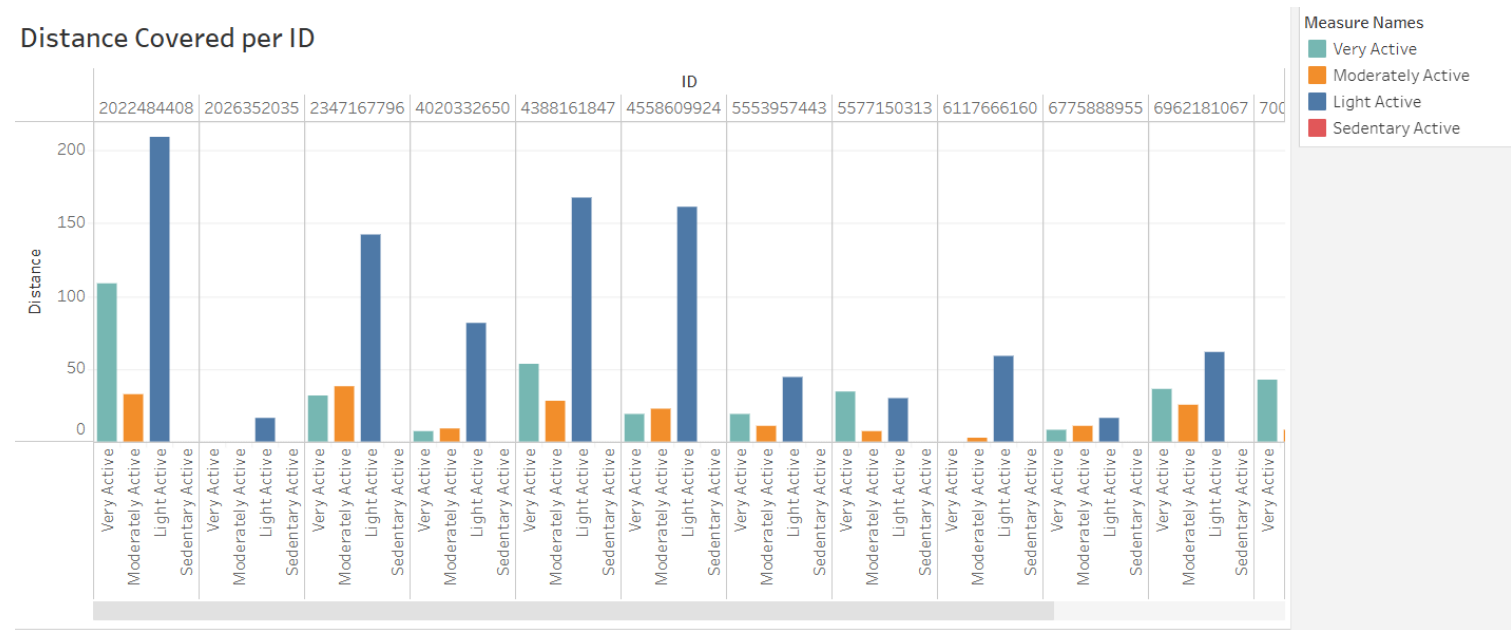
- **Dataset:** FitBit Fitness Tracker Data
- 30 users with daily & hourly tracking
- **Data includes:**
  - Steps
  - Calories
  - Intensity
  - Heart rate
  - Sleep
- **Focus of Analysis:**
  - Daily activity patterns
  - Relationships between health metrics
  - Weekly behavior trends

| Sum of Steps | Time  |       |      |      |      |       |       |        |        |        |        |        |        |        |        |        |        |        |        |        |        |       |       |       |
|--------------|-------|-------|------|------|------|-------|-------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|-------|-------|-------|
| Week Days    | 12 AM | 1 AM  | 2 AM | 3 AM | 4 AM | 5 AM  | 6 AM  | 7 AM   | 8 AM   | 9 AM   | 10 AM  | 11 AM  | 12 PM  | 1 PM   | 2 PM   | 3 PM   | 4 PM   | 5 PM   | 6 PM   | 7 PM   | 8 PM   | 9 PM  | 10 PM | 11 PM |
| Monday       | 9267  | 4172  | 2105 | 1600 | 4037 | 10558 | 41721 | 85245  | 117182 | 118439 | 135286 | 119098 | 149489 | 135368 | 117211 | 89699  | 124886 | 122214 | 155853 | 170691 | 117398 | 66138 | 38140 | 22148 |
| Tuesday      | 8154  | 6173  | 3020 | 2428 | 5439 | 17624 | 56048 | 102014 | 117607 | 106594 | 132407 | 115087 | 170612 | 138531 | 110684 | 88639  | 125818 | 144280 | 179272 | 150068 | 115122 | 98088 | 73294 | 31892 |
| Wednesday    | 12717 | 5023  | 3477 | 1076 | 3048 | 8975  | 46917 | 82770  | 124533 | 124538 | 101264 | 125078 | 132895 | 122867 | 143518 | 107902 | 120570 | 173587 | 196539 | 176818 | 90448  | 66825 | 52179 | 29987 |
| Thursday     | 8288  | 4199  | 3044 | 1455 | 2163 | 9951  | 54041 | 86518  | 113032 | 132582 | 139937 | 118236 | 152296 | 123386 | 137562 | 100095 | 122338 | 130659 | 127694 | 147202 | 98236  | 88989 | 53291 | 27718 |
| Friday       | 11836 | 5391  | 3540 | 2072 | 3215 | 12171 | 49378 | 84419  | 114830 | 115434 | 88196  | 110107 | 120872 | 125697 | 125385 | 83917  | 117871 | 115590 | 143586 | 145486 | 97255  | 67814 | 56022 | 31038 |
| Saturday     | 14561 | 6350  | 4018 | 3095 | 1958 | 3704  | 20620 | 61232  | 92187  | 125679 | 136242 | 161962 | 166368 | 166038 | 176323 | 141650 | 158001 | 129963 | 133721 | 135930 | 106762 | 80920 | 69903 | 36118 |
| Sunday       | 19081 | 11016 | 7281 | 1516 | 1591 | 4460  | 17382 | 43326  | 84603  | 109284 | 140795 | 125356 | 132177 | 139384 | 159358 | 148397 | 126285 | 133160 | 108290 | 127533 | 91141  | 68751 | 43566 | 33052 |

# Analysis Findings

## Daily Activity Patterns

- Most activity occurs between 7 AM – 7 PM
- Peak times:
  - 12 PM (midday)
  - 7 PM (evening)
- Intensity & heart rate follow the same pattern
- Majority of users are **lightly active**
- Average sleep: ~7 hours/day
- **Key Insight:**  
Users are most active during structured daytime hours.



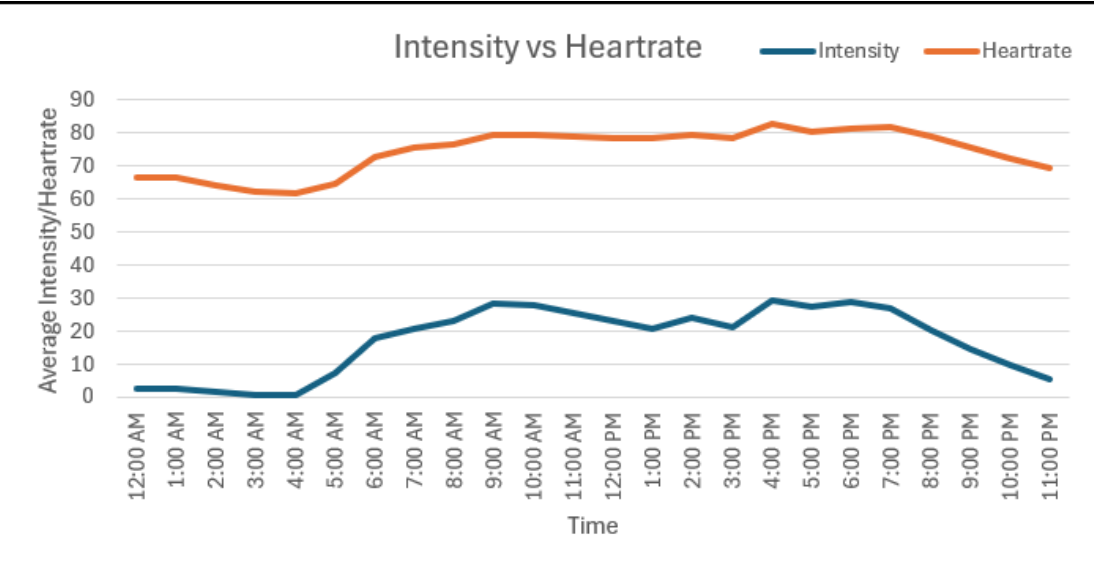
# Analysis Findings

## Key Relationships & Trends

- Positive relationships:
  - Steps  $\uparrow$   $\rightarrow$  Calories  $\uparrow$
  - Intensity  $\uparrow$   $\rightarrow$  Heart Rate  $\uparrow$
  - All metrics are interconnected: increases in activity (steps and intensity) directly lead to higher calorie burn and heart rate.
- **Saturday = highest activity day**
- **Key Insight:**

Users show consistent patterns but mostly low-intensity activity.

| Week      |             |                |                 |                |          |
|-----------|-------------|----------------|-----------------|----------------|----------|
| Day       | Total Steps | Total Distance | Total Intensity | Avg. Heartrate | Calories |
| Saturday  | 416,671     | 294            | 15,045          | 78             | 105,412  |
| Monday    | 382,234     | 272            | 13,792          | 78             | 97,089   |
| Friday    | 373,708     | 272            | 13,975          | 78             | 103,786  |
| Wednesday | 368,294     | 264            | 13,188          | 76             | 90,657   |
| Tuesday   | 357,198     | 256            | 13,865          | 77             | 98,143   |
| Sunday    | 330,479     | 238            | 12,480          | 78             | 95,081   |
| Thursday  | 323,748     | 228            | 11,838          | 78             | 84,083   |



# Recommendations



Target users during **peak activity times**



Encourage transition from **light to active lifestyle**



Focus marketing on **weekend engagement**

# Actionable Steps



- **Implement time-based notifications:**

Send notifications and campaigns during Midday and Evening

- **Leverage Weekend Engagement:**

Launch Weekend challenges & Saturday campaigns

- **Promote Active Lifestyle:**

Encourage users to move from Light activity → Moderate/High activity



# Appendix

For more detailed analysis, visualizations, and insights:

- Refer to the **Excel datasets**.
- Explore additional **charts and visualizations in Tableau**.

These resources provide deeper context beyond the main presentation.